(Please write your Enrollment Number) Enrollment No.________________

End-Term Examination
(NEP)(SUBJECTIVE TYPE)(Offline)
Course Name:<B.Tech. - >
Semester:<1^{st}> 2024-25

Subject Code: BAS 101	**Subject: Applied Mathematics**
Time :3 Hours	**Maximum Marks : 60**

Note : Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	**CO Mapping**
	(a) Verify if the following matrix is orthogonal and hence, find its inverse: $A = \begin{bmatrix} \cos\emptyset & -\sin\emptyset & 0 \\ \sin\emptyset & \cos\emptyset & 0 \\ 0 & 0 & 1 \end{bmatrix}$		CO1
	(b) Determine the value(s) of α so that rank of A is 3, where $A = \begin{bmatrix} 1 & 1 & -1 & 0 \\ 4 & 4 & -3 & 1 \\ \alpha & 2 & 2 & 2 \\ 9 & 9 & \alpha & 3 \end{bmatrix}$		CO1
	(c) Find the n^{th} derivative of $e^{-3x}cos^3x$.		CO2
	(d) Prove that the function $\sqrt{\lvert xy\rvert}$ is continuous at origin.		CO2
	(e) Evaluate $\int_0^\infty \int_0^x xe^{-\frac{x^2}{y}}\,dy\,dx$ by changing the order of integration.		CO3
	(f) Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{a\sin\theta} \int_0^{\frac{(a^2-r^2)}{a}} r\,dz\,dr\,d\theta$.		CO3
	(g) Prove that $\nabla r^n = nr^{n-2}\bar{r}$, where $\bar{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}, r = \lvert\bar{r}\rvert$		CO4
	(h) Find the equation of tangent plane and normal vector to the level surface $f(x,y,z) = x^2 + y^2 + z - 9 = 0$ at the point $P_0(1,2,4)$.		CO4
	UNIT-I		**CO Mapping**
Q2	(a) Check whether the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$ is diagonalizable. If yes, then find out the modal matrix *P*. If not, then give reasons. (b) Show that the vectors $x_1 = (1,2,4), x_2 = (2,-1,3)$, $x_3 = (0,1,2)$, $x_4 = (-3,7,2)$, are linearly dependent and find the relation between them.	**(5*2=10)**	CO1
Q3	(a) Find the non-singular matrices P and Q such that PAQ is in the normal form and hence, find the rank of the matrix A, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$ (b) Determine the values for a and b for which the system $x + 2y + 3z = 6$, $x + 3y + 5z = 9$, $2x + 5y + az = b$	**(5*2=10)**	CO1

	has (i) no solution (ii) unique solution (iii) infinite number of solutions.		
	UNIT-II		**CO Mapping**
Q4	(a) If $y = e^{\tan^{-1}x}$, prove that $(1+x^2)y_{n+2} + [2(n+1)x - 1]y_{n+1} + n(n+1)y_n = 0.$ (b) Expand $e^x \log(1+y)$ in powers of x and y up to terms of third degree.	**(5*2=10)**	CO2
Q5	(a) Find the extreme values of $$f(x,y) = 3y^2 - 2y^3 - 3x^2 + 6xy.$$ (b) Apply Taylor's series to calculate the value of $f\left(\frac{11}{10}\right)$, where $f(x) = x^3 + 3x^2 + 15x - 10$.	**(5*2=10)**	CO2
	UNIT-III		**CO Mapping**
Q6	(a) Trace the curve $r = a\cos 2\theta$. (b) Compute $\iint_D \left(\frac{x-y}{x+y}\right)^4 dy\,dx$, where D is the region bounded by the lines $x + y = 1$ and the coordinate axes by taking the transformation $u = x - y$, $v = x + y$.	**(5*2=10)**	CO3
Q7	(a) Find the volume of the tetrahedron bounded by the planes $2x + y + 3z = 6,\ x = 0,\ y = 0\ and\ z = 0$. (b) Trace the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$.	**(5*2=10)**	CO3
	UNIT-IV		**CO Mapping**
Q8	(a) Determine whether $\vec{F} = (y^2\cos x + z^3)\hat{\imath} + (2y\sin x - 4)\hat{\jmath} + (3xz^2 + 2)\hat{k}$ Is a conservative vector field. If so, find the scalar potential. Also, compute the work done in moving the particle from (0,1,-1) to $(\frac{\pi}{2}, -1, 2)$. (b) Verify Gauss's Divergence theorem for the vector $\vec{F} = y\hat{\imath} + x\hat{\jmath} + z^3\hat{k}$ taken over the cylindrical region $x^2 + y^2 = 9; z = 0, z = 6$.	**(5*2=10)**	CO4
Q9	(a) Verify the Green's theorem for $$\int_C e^{-x}\sin y\,dx + e^{-x}\cos y\,dy$$ where C is the rectangle with the vertices $(0,0), (\pi,0), \left(\pi,\frac{\pi}{2}\right), \left(0,\frac{\pi}{2}\right)$ (b) Verify Stokes theorem for $\bar{F} = xz\hat{\imath} + y\hat{\jmath} + xy^2\hat{k}$ where S is the surface of the region bounded by $y = 0, z = 0$, and $4x + y + 2z = 4$ which is not included in the yz-plane.	**(5*2=10)**	CO4